

Project #3 Drawing and Tolerances

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Overview

This project focused on producing professional engineering drawings and applying ISO limits and fits to a hand-cranked Ferris wheel mechanism modeled in SOLIDWORKS. The three selected parts were the Ferris wheel, axle shaft, and A-frame support. The parts were modeled from the physical mechanism, assembled with the required mates, and checked in three mechanism configurations. An exploded assembly drawing, bill of materials, balloons, and individual part drawings were also created.

The critical interfaces are the axle shaft passing through the Ferris wheel hub and through the A-frame support. Because the axle must rotate freely while remaining accurately located, a close-running clearance fit was selected. The selected ISO fit is H8/f7. The resulting limits were converted to bilateral plus/minus values for the SOLIDWORKS drawings, and the corresponding maximum clearance and allowance were calculated for both interfaces.

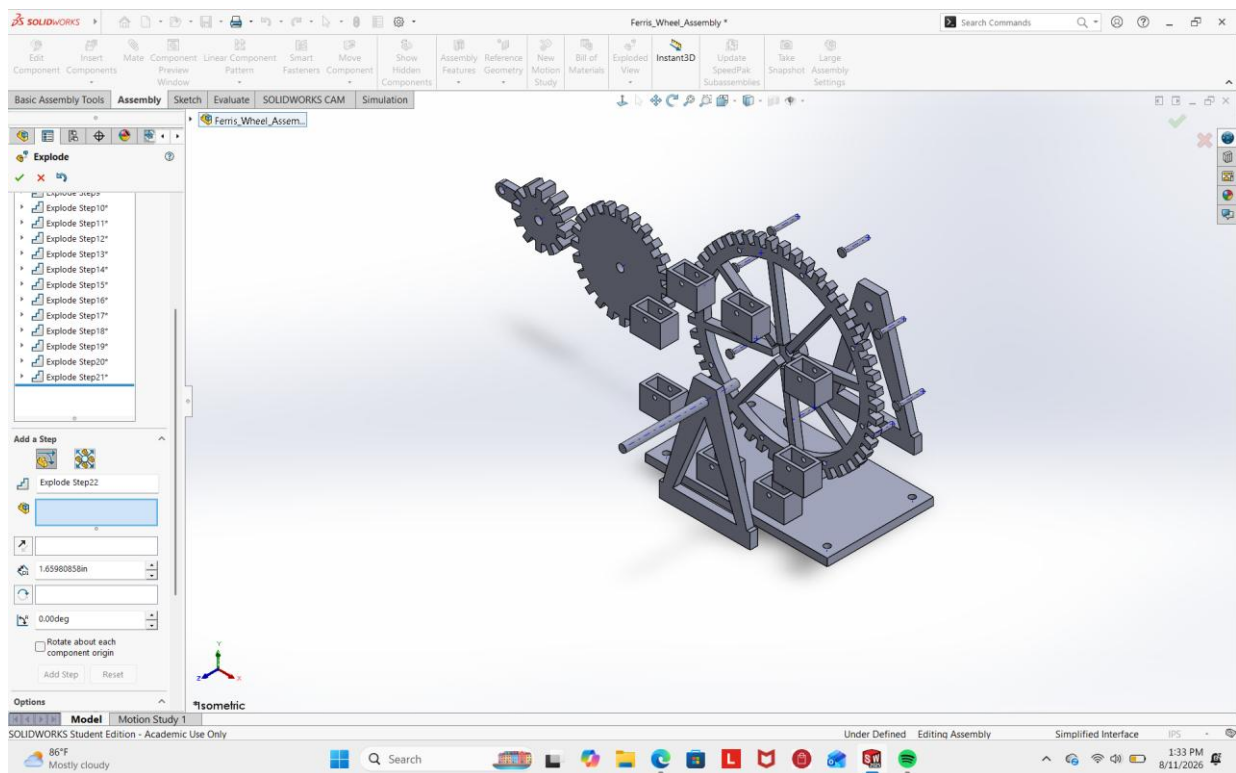


Figure 1. Exploded view of the Ferris wheel assembly showing the major parts separated from the assembled position.

Part Files

The three selected parts are the A-frame support, axle shaft, and Ferris wheel. The screenshots below show the part geometry and expanded feature trees used to construct each model. The A-

frame sketch shown is fully defined. The final drawings identify 6061 Aluminum Alloy as the assigned material for the parts.

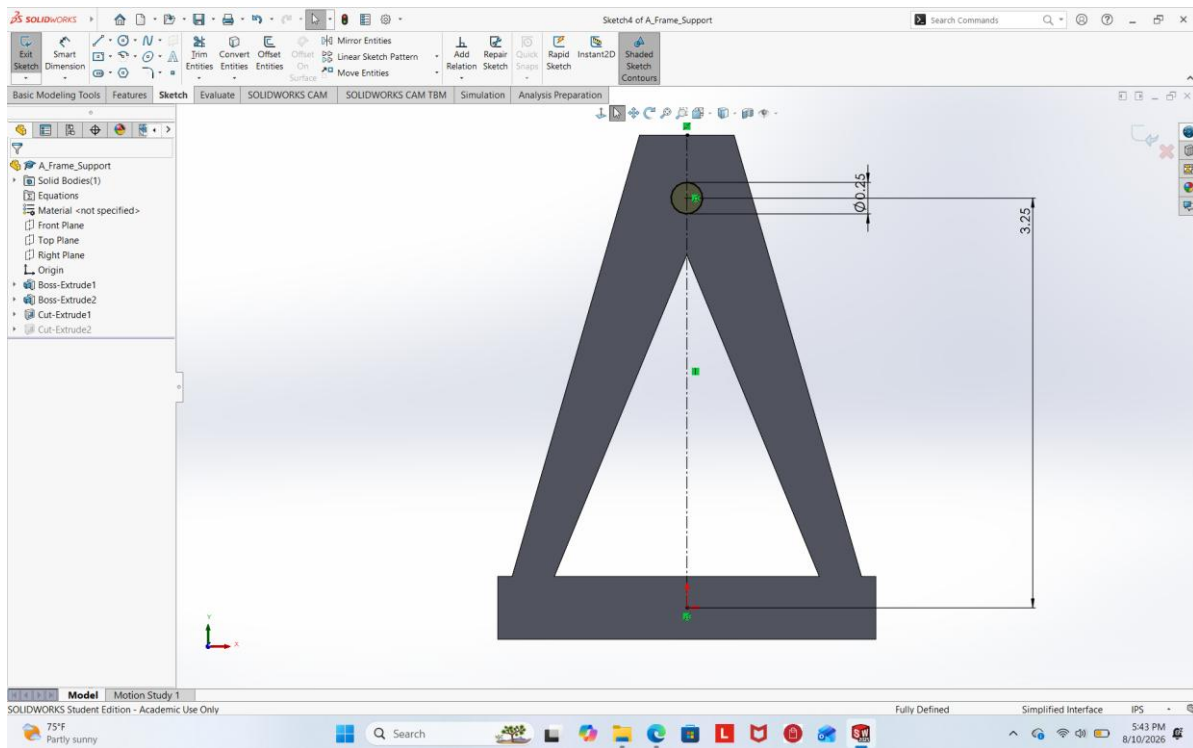


Figure 2. A-frame support part file with the feature tree visible and the axle-hole sketch fully defined.

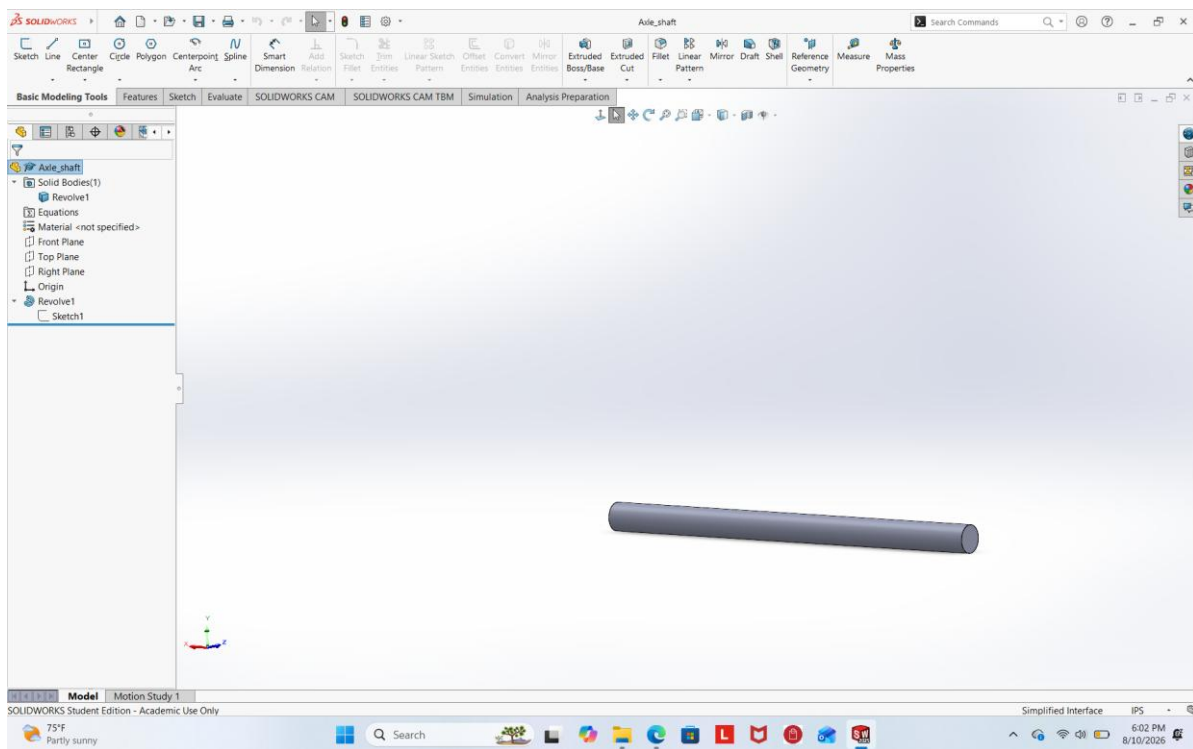


Figure 3. Axle shaft part file with the revolve feature and sketch visible in the feature tree.

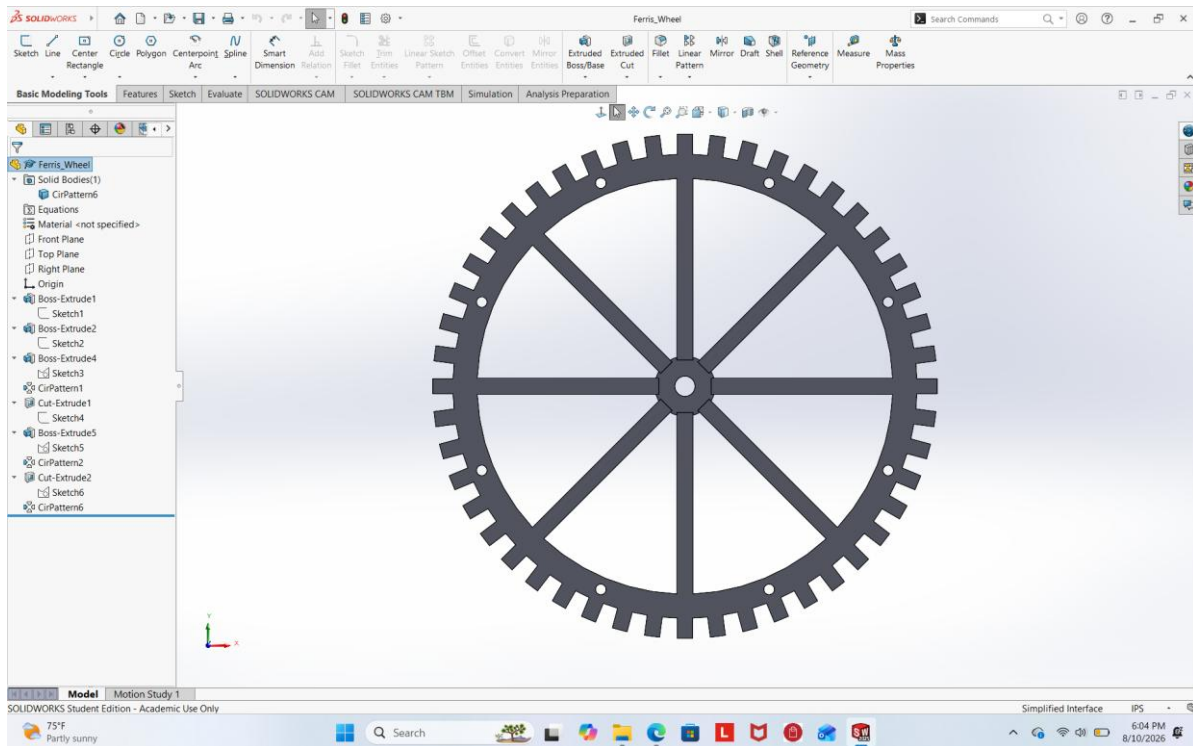


Figure 4. Ferris wheel part file with the circular patterns, cut features, and wheel geometry visible.

Assembly File

The completed assembly was saved in three configurations representing different crank and wheel positions. These configurations were created so the mechanism could be documented at multiple points in its motion. The screenshots below show the three saved states of the full assembly.

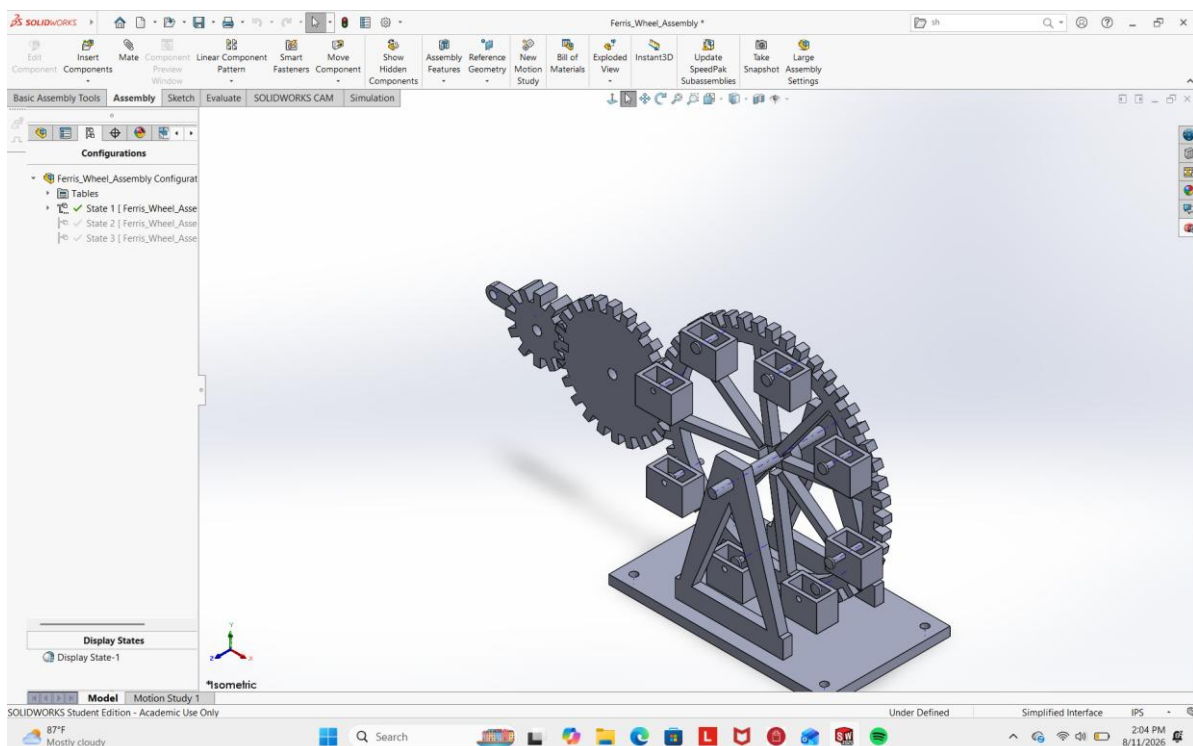


Figure 5. Ferris wheel assembly, State 1.

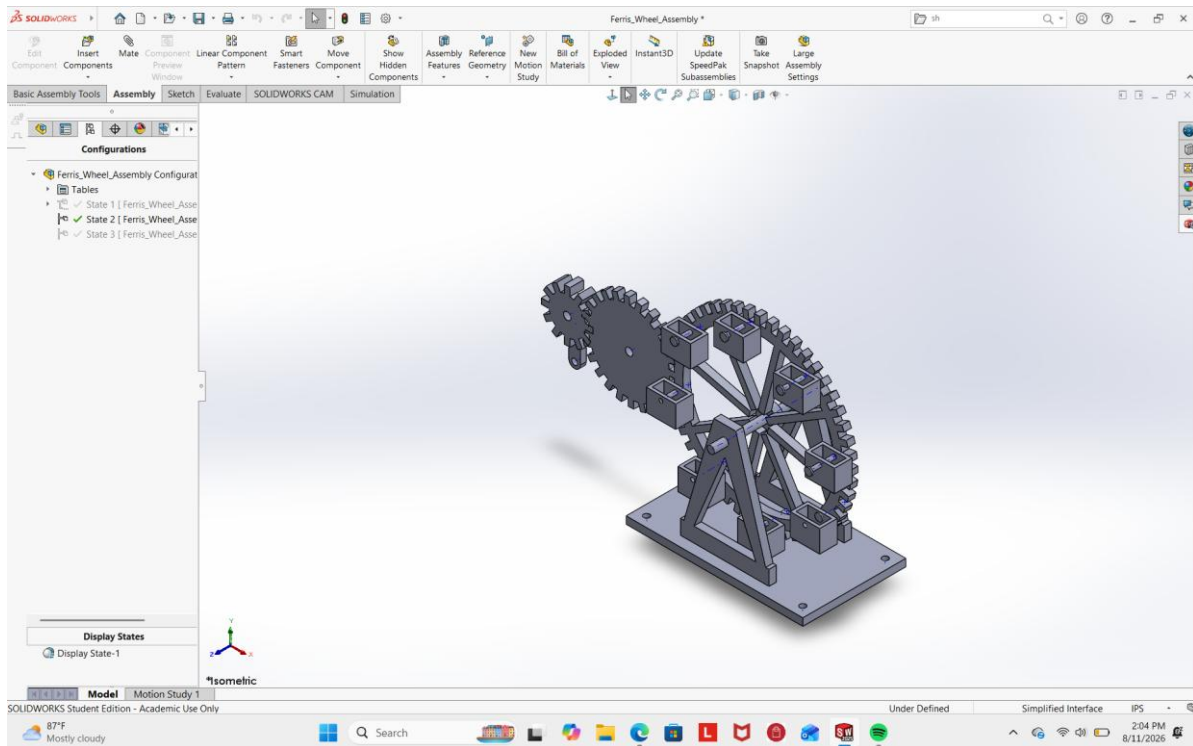


Figure 6. Ferris wheel assembly, State 2.

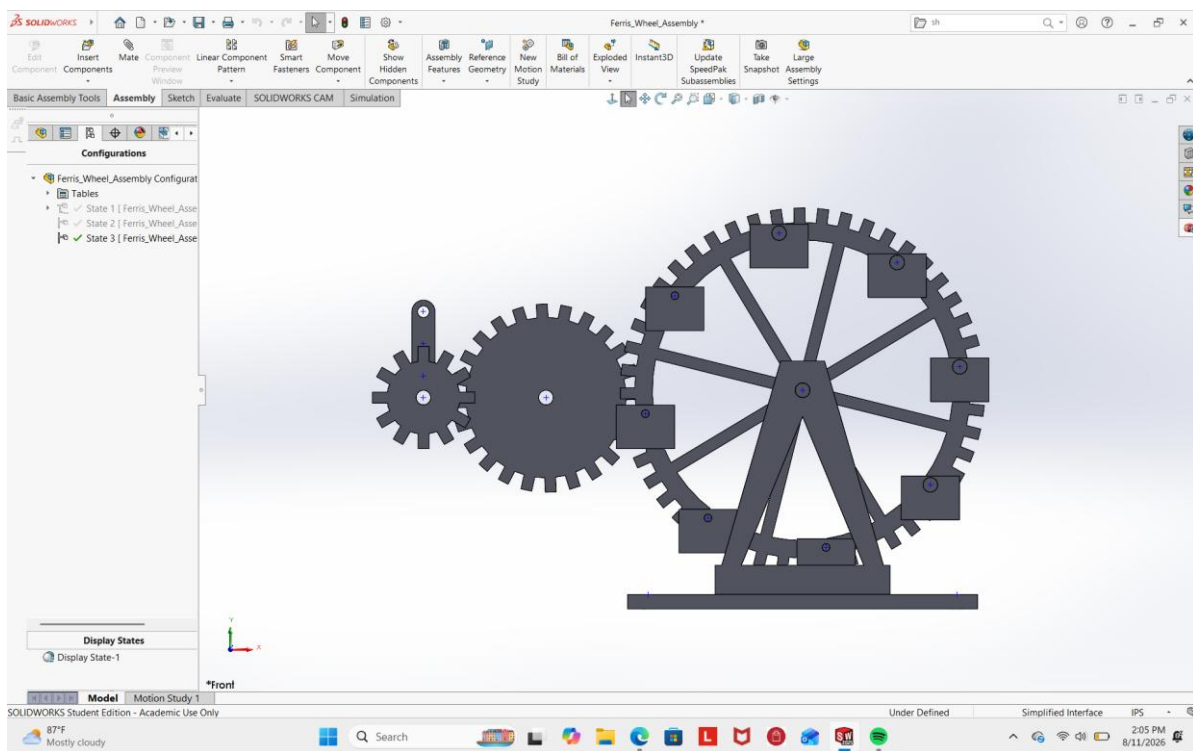


Figure 7. Ferris wheel assembly, State 3.

An exploded assembly drawing was created on the course drawing template and includes a bill of materials and item balloons. This drawing documents all eight unique component types in the assembly, including the repeated gondolas and pivot pins.

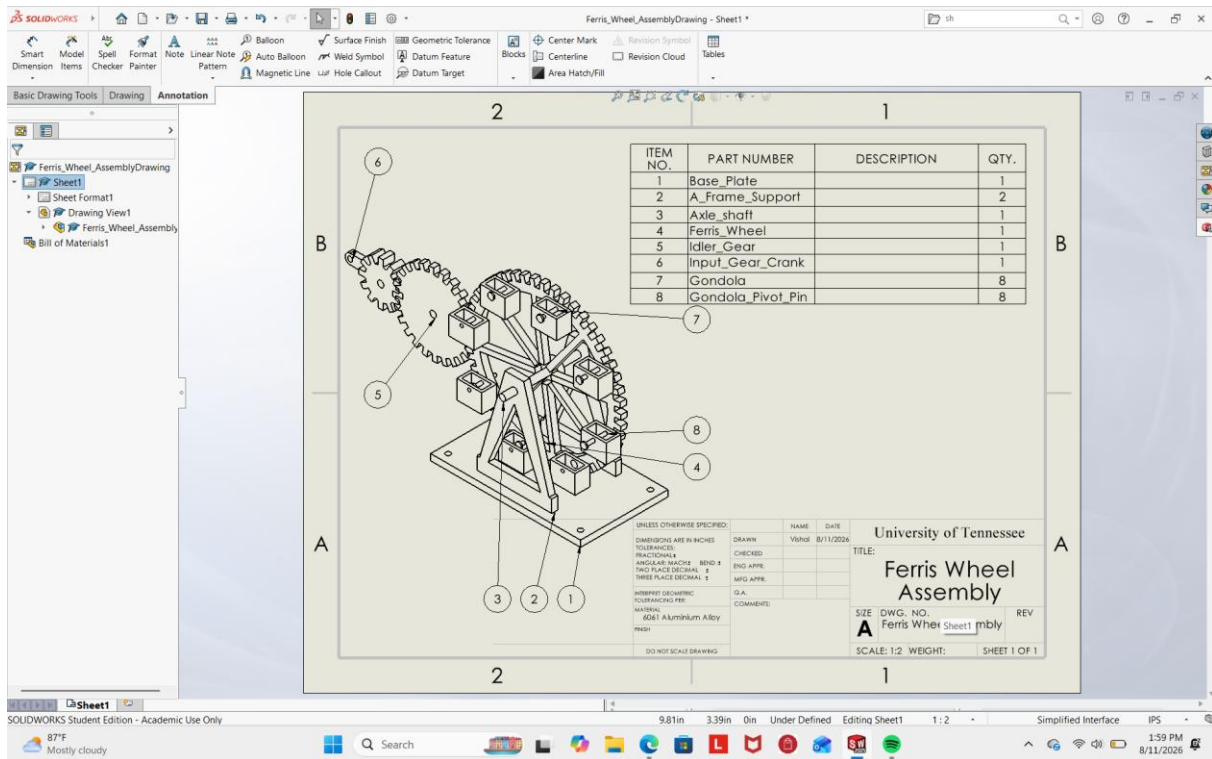


Figure 8. Assembly drawing with bill of materials and item balloons.

Fit Selection and Tolerances

A close-running clearance fit was selected for both the Ferris wheel/axle interface and the A-frame/axle interface. The ISO fit callout is H8/f7. This fit was selected because the axle must rotate freely through the wheel and supports without binding, while the clearance must remain small enough to keep the wheel properly located. The limits used in the drawings are inclusive of the intended 0.250 in basic size. Figure 9 shows every calculation step for the upper limit, lower limit, allowance, and maximum clearance for both interfaces.

Project 3 – Fit Calculations

Ferris Wheel Assembly

1. Interface 1: Axle Shaft ↔ Ferris Wheel

Hole fit: H8 | Shaft fit: f7

Ferris-wheel hole, H8

$$D_{max} = 0.25045 + 0.00045 = 0.25090 \text{ in}$$

$$D_{min} = 0.25045 - 0.00045 = 0.25000 \text{ in}$$

Axle shaft, f7

$$d_{max} = 0.2492 + 0.0003 = 0.24950 \text{ in}$$

$$d_{min} = 0.2492 - 0.0003 = 0.24890 \text{ in}$$

Allowance (minimum clearance)

$$A = D_{min} - d_{max}$$

$$A = 0.25000 - 0.24950$$

$$A = 0.00050 \text{ in}$$

Maximum clearance

$$C_{max} = D_{max} - d_{min}$$

$$C_{max} = 0.25090 - 0.24890$$

$$C_{max} = 0.00200 \text{ in}$$

2. Interface 2: Axle Shaft ↔ A-Frame Support

Hole fit: H8 | Shaft fit: f7

A-frame hole, H8

$$D_{max} = 0.25045 + 0.00045 = 0.25090 \text{ in}$$

$$D_{min} = 0.25045 - 0.00045 = 0.25000 \text{ in}$$

Axle shaft, f7

$$d_{max} = 0.2492 + 0.0003 = 0.24950 \text{ in}$$

$$d_{min} = 0.2492 - 0.0003 = 0.24890 \text{ in}$$

Allowance (minimum clearance)

$$A = D_{min} - d_{max}$$

$$A = 0.25000 - 0.24950$$

$$A = 0.00050 \text{ in}$$

Maximum clearance

$$C_{max} = D_{max} - d_{min}$$

$$C_{max} = 0.25090 - 0.24890$$

$$C_{max} = 0.00200 \text{ in}$$

Final Results

Interface	Quantity	Result
Ferris Wheel / Axle	Allowance	0.00050 in
Ferris Wheel / Axle	Maximum Clearance	0.00200 in
A-Frame / Axle	Allowance	0.00050 in
A-Frame / Axle	Maximum Clearance	0.00200 in

Justification: The selected H8/f7 fit provides a close-running clearance fit. The axle can rotate smoothly through the Ferris wheel and A-frame support while remaining properly located. Because the minimum clearance is positive, there is no interference; because the maximum clearance is small, the fit remains controlled and suitable for this assembly.

Figure 9. Complete H8/f7 limit, allowance, and maximum-clearance calculations for both fitted interfaces.

Table 2. Calculated Tolerances

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	Major Interface: Ferris Wheel Center		Hole / Axle Shaft
	Close-running clearance fit		
		H8/f7	
	Base	Upper Limit	Lower Limit
Part 1 - Ferris Wheel Hole (H8)	0.25000 in	0.25090 in	0.25000 in
Part 2 - Axle Shaft (f7)	0.25000 in	0.24950 in	0.24890 in
Clearance	Maximum = 0.00200 in	Maximum = 0.00200 in	Maximum = 0.00200 in
Allowance	Minimum clearance = 0.00050 in	Minimum clearance = 0.00050 in	Minimum clearance = 0.00050 in

	Minor Interface: A-Frame Support		Hole / Axle Shaft
	Close-running clearance fit		
		H8/f7	
	Base	Upper Limit	Lower Limit
Part 1 - A-Frame Hole (H8)	0.25000 in	0.25090 in	0.25000 in
Part 2 - Axle Shaft (f7)	0.25000 in	0.24950 in	0.24890 in
Clearance	Maximum = 0.00200 in	Maximum = 0.00200 in	Maximum = 0.00200 in
Allowance	Minimum clearance = 0.00050 in	Minimum clearance = 0.00050 in	Minimum clearance = 0.00050 in

Drawings

The following three drawings document the axle shaft, A-frame support, and Ferris wheel. The axle-shaft drawing contains the shaft tolerance, while the A-frame and Ferris-wheel drawings contain the corresponding H8 hole tolerances. Each drawing uses the course title block and inch units.

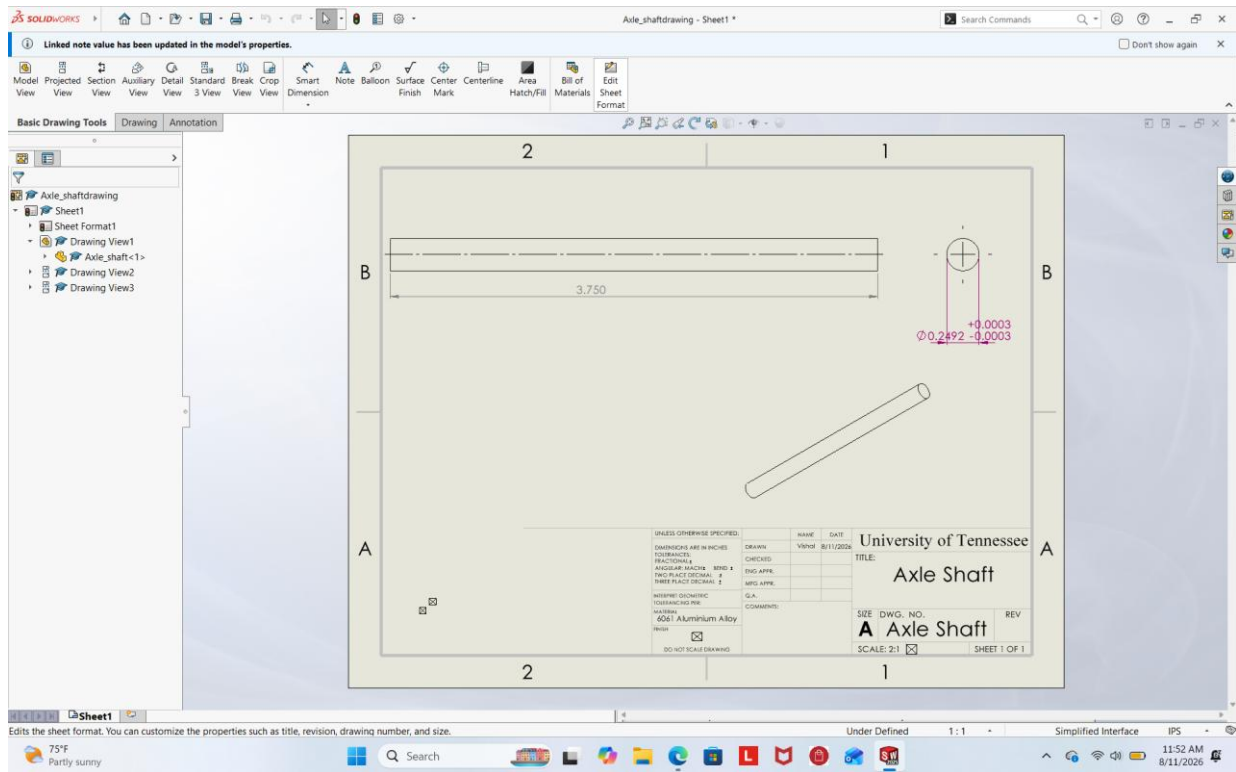


Figure 10. Axle shaft drawing with the f7 shaft tolerance.

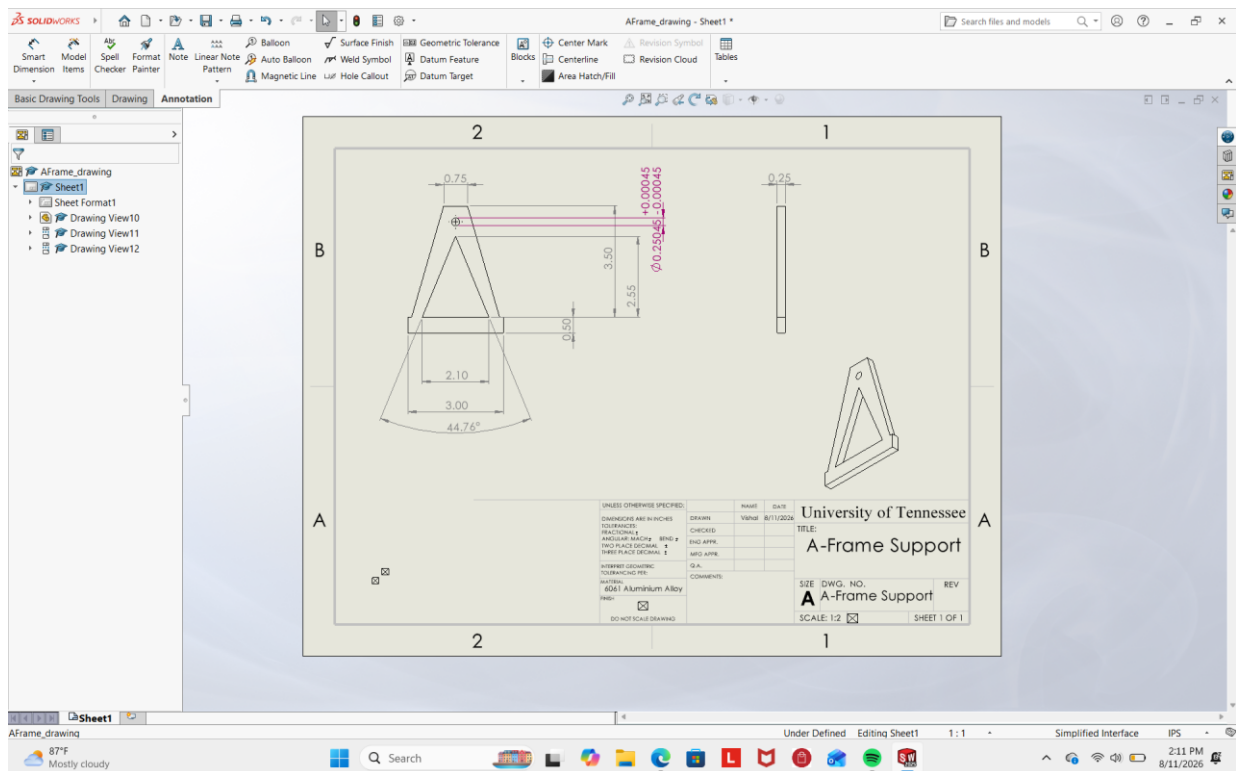


Figure 11. A-frame support drawing with the H8 axle-hole tolerance.

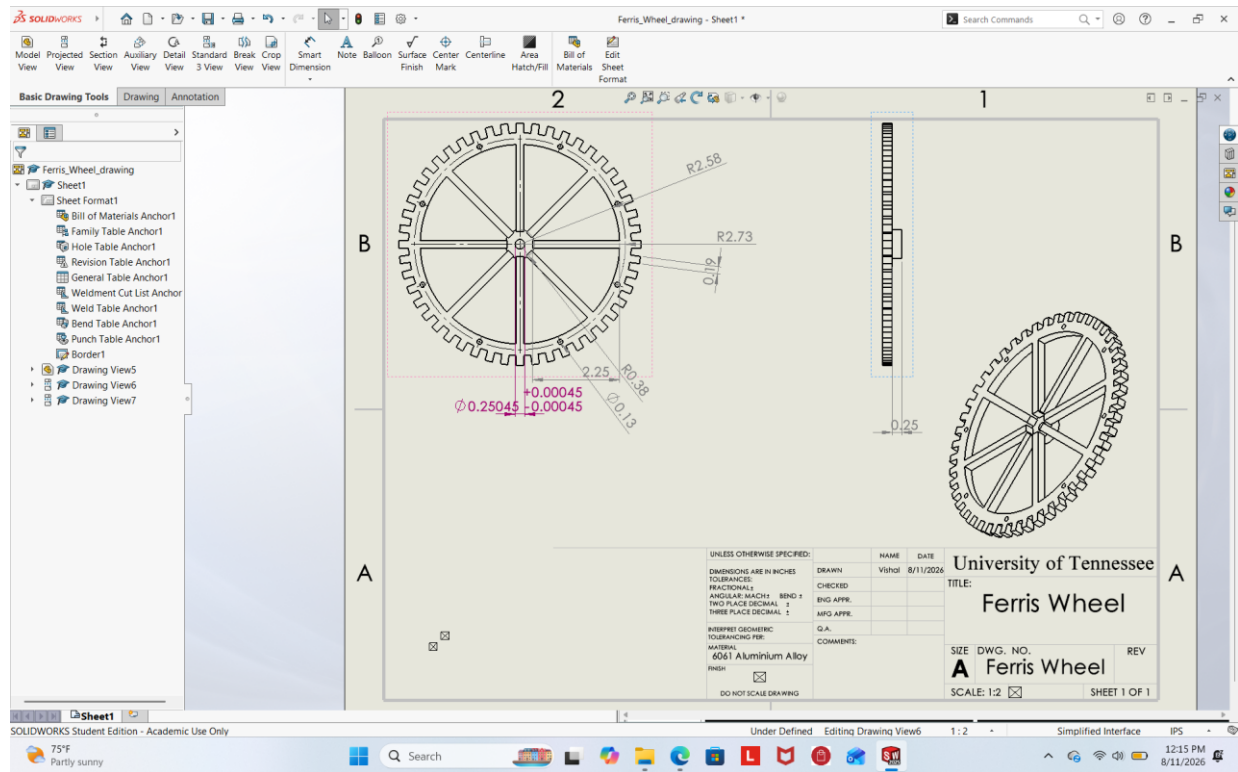


Figure 12. Ferris wheel drawing with the H8 center-hole tolerance.

Conclusion

This project connected physical measurement, CAD modeling, assembly constraints, ISO limits and fits, and professional drawing practices. I modeled and assembled the Ferris wheel mechanism, documented three mechanism configurations, created an exploded assembly drawing with a bill of materials, and produced fully dimensioned drawings for the three selected parts. The H8/f7 close-running clearance fit was selected because it provides guaranteed positive clearance for rotation while keeping the axle accurately located. The project reinforced how tolerance selection directly affects whether separately manufactured parts will assemble and function as intended, and it showed how calculated limits must be carried consistently from the engineering analysis into the final drawings.